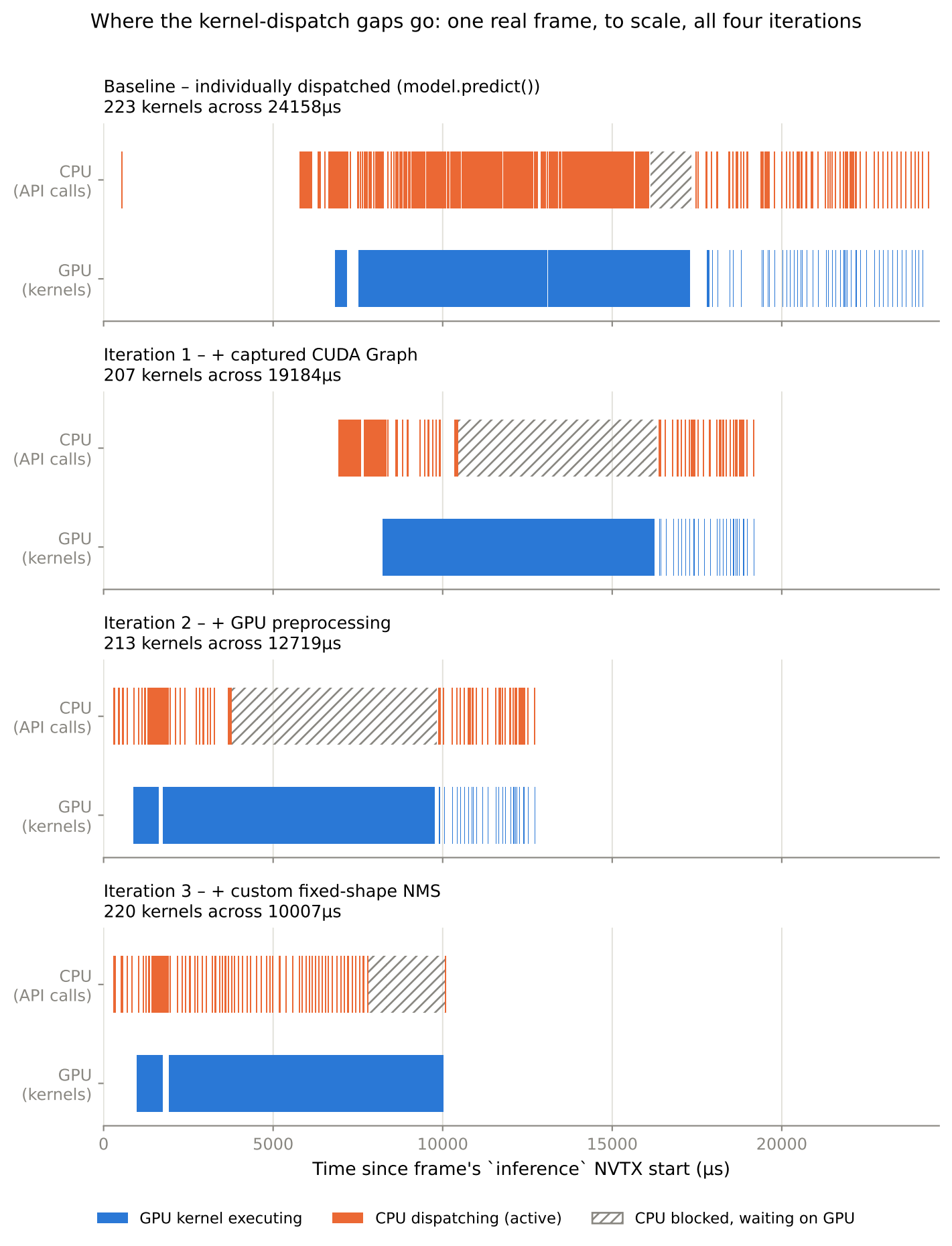
Where the kernel-dispatch gaps go: one real frame, to scale, all four iterations
Baseline – individually dispatched (model.predict())
223 kernels across 24158µs
CPU (API calls)
GPU (kernels)
Iteration 1 – + captured CUDA Graph
207 kernels across 19184µs
CPU (API calls)
GPU (kernels)
Iteration 2 – + GPU preprocessing
213 kernels across 12719µs
CPU (API calls)
GPU (kernels)
Iteration 3 – + custom fixed-shape NMS
220 kernels across 10007µs
CPU (API calls)
GPU (kernels)
0
5000
10000
15000
20000
Time since frame's `inference` NVTX start (µs)
GPU kernel executing
CPU dispatching (active)
CPU blocked, waiting on GPU